Miaohui Shi

施妙辉

Algorithm Engineer | High-performance Computing & Robotics

Taizhou, Zhejiang, China | shimiaohui0426.github.io
GitHub / ShiMiaohui0426 • LinkedIn / Miaohui Shi

Algorithm engineer taking technical ownership of core computational lithography software: boundary-focused algorithms, C++ performance optimization and recoverable multi-node pipelines. Waseda M.S.; research in human-robot interaction and tactile sensing.

44 → 24 min
4-pass processing time
About 45% less • same 6 nodes / data

< 120 GB
Observed memory use
Block processing • OOM resolved

4 / 4
Subsystems with unified logs
FastAPI • searchable task history

PROFESSIONAL EXPERIENCE

Taizhou Optoelectronics Industry Innovation Center

Sep 2024-present

Algorithm Engineer | Core computation technical lead (project role)

- Guide core technical decisions and task coordination in a six-person team. Building on colleagues' Python prototype, proposed boundary-focused rasterization and led C++ / OpenMP and multi-node implementation.
- Loop fusion and flame-graph-guided Clipper2 optimization cut 4-pass runtime from 44 to 24 min (about 45%) on the same six nodes and test data. The 24-min run includes correction and format conversion.
- Block processing, preallocation and buffer reuse resolved OOM on 256 GB nodes; observed memory use below 120 GB. Overlapped I/O with computation and reduced intermediate data.
- Proposed an online equipment data pipeline with shared-filesystem readiness signals and backpressure. Persisted stripe-level state enables computational checkpoint recovery, verified by interruption tests.
- Built FastAPI logging for all four equipment subsystems. Maintain startup / operation guides for remote support; collaborate with testers on manual regression using a fixed GDS library.

Hesai Technology / Sharpa-related R&D

Apr-Aug 2024

Algorithm Development

- At Hesai Technology, developed LiDAR functional-degradation algorithms and trained noise-segmentation models; also contributed to Sharpa-related tactile-sensor R&D.
- Designed and trained vision-based tactile-sensor mechanical models; developed software on Ubuntu.

EDUCATION

Waseda University

M.S., Mechanical Engineering
2022-Mar 2024 • Japan

Huazhong University of Science and Technology

B.S., Mechanical Science & Engineering
2017-2021 • China

TECHNICAL SKILLS

Performance & engineering
C++ • OpenMP • Python • Linux • profiling • memory / I/O optimization

Vision & learning
YOLO • segmentation • anomaly detection • model training • vision-based tactile sensing

Robotics & interaction
ROS • Unity • HRI experiments • PASAT / NASA-TLX • behavioral analysis

Software & interfaces
FastAPI • HTTP APIs • logging • checkpoint recovery • C# / Qt • TwinCAT / PLC

Miaohui Shi

SELECTED RESEARCH

Multi-robot interaction, attention and automation | CASE 2024

Independently developed the experimental system (excluding the mechanical-head module), designed and conducted the experiments, and completed all data analysis. Adapted the PASAT auditory-addition task to a dual-task study involving three robotic arms, with NASA-TLX and user-experience measures.

High-speed vision-based tactile sensing | ICAT-EGVE 2024

Independently developed algorithmic principles, implementation, performance tuning and the front-end UI. The method combines local sparse marker tracking with force estimation using an asymmetric stiffness-coefficient matrix. The paper reports 601.25 Hz sensor-system force acquisition; this is not a standalone algorithm benchmark.

PUBLICATIONS

2024 **High-Speed Vision-Based Haptic Sensor for Robotic Dermatological Palpation: Force Sensing Method Using Asymmetric Stiffness Coefficient Matrix**
Fumihiro Kato, **Miaohui Shi**, Kaito Kamishima, Hiroyasu Iwata
ICAT-EGVE 2024

2024 **The Role of automation and User Experience in Multi Robot Systems**
Kouta Suzuki, **Miaohui Shi**, Vimolmongkolporn Vitvasin, Yukiko Iwasaki, Hiroyasu Iwata
IEEE CASE 2024 · pp. 3818-3825

2023 **Balancing Real-world Interaction and VR Immersion with AI Vision Robotic Arm**
Xiaoran Yang, Yang Zhan, Yukiko Iwasaki, **Miaohui Shi**, Shijie Tang, Hiroyasu Iwata
IEEE ICMA 2023 · pp. 2051-2057

2021 **基于TwinCAT的机器人实验平台设计**
黄弢, **施妙辉**, 汪迪, 王峻峰
实验技术与管理 · 38(12):197-202+207

2021 **A Remote Experiment Platform based on HTML5 for Industrial Robots**
Tao Huang, **Miaohui Shi**
MED 2021 · pp. 920-923

PATENTS

CN119987127B | Subpixel image processing based on grid-partitioned mask regions
Co-inventor (fourth listed) | Granted 20 March 2026
Full patent record / 8 applications

EARLIER ACADEMIC EXPERIENCE

Oct-Dec 2022 | Learning Assistant · Waseda University
Dec 2019-Feb 2022 | Research Assistant · Huazhong University of Science and Technology